

Trainings ▾

Preparatory Steps

WARNING

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

CAUTION

Use caution when disconnecting or removing fuel lines, replacing filters, and priming the fuel system that fuel is not spilled or drained into the bilge area. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be discarded in accordance with local environmental regulations.

CAUTION

Zinc plugs expand and can break off during removal. Inspect the zinc plug to make sure it is in one piece. If not, it must be replaced with a new zinc plug, and the broken pieces must be retrieved from the aftercooler to prevent damage to components downstream in the sea water system.

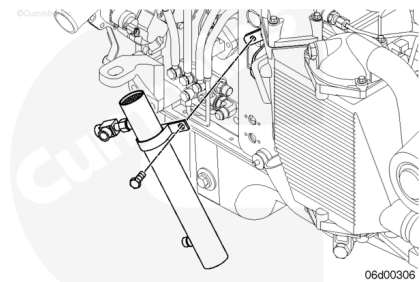
- Shut off the sea water supply valve(s). See equipment manufacturer service information.
- Drain the sea water system by removing a zinc plug from the aftercooler. Refer to Procedure 010-005 in Section 10. (</qs3/pubsys2/xml/en/procedures/40/40-010-005-tr.html>)
- Shut off the fuel supply and return valves to the engine. See equipment manufacturer service information.
- Disconnect the fuel drain line from the combined fuel manifold to the fuel cooler at the fuel cooler connection.
- Disconnect the fuel drain line from the fuel cooler to the fuel tanks.

Remove

Remove the hose clamps at the top and bottom of the fuel cooler.

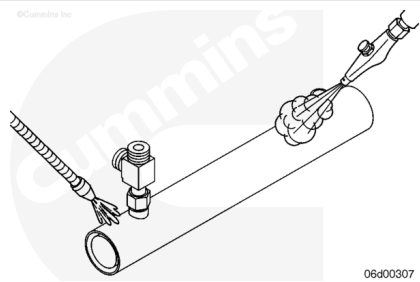
Remove the fuel cooler mounting clamp capscrew from the fuel cooler mounting bracket.

Remove the fuel cooler.



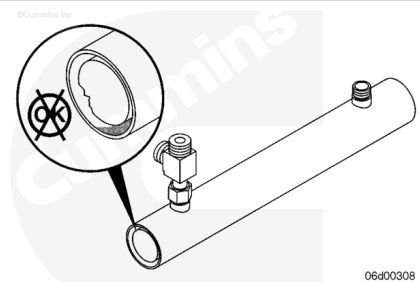
Clean and Inspect for Reuse

Clean the hose sealing surfaces of the fuel cooler.



Inspect the core area of the fuel cooler for debris. Clean if required.

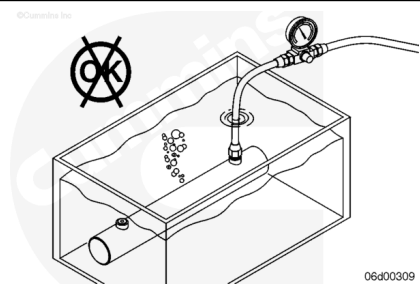
Inspect the core for damage to the inner core surfaces.



Pressure test the fuel side of the fuel cooler.

Install a plug in one end of the cooler.

Install fittings and apply shop air pressure.



Measurements		
	kpa	psi
Air Pressure	172	25

Place the cooler in a tank of water. Check for leaks.

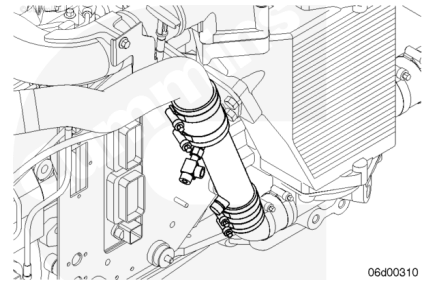
Install

Install the fuel cooler to the fuel cooler mounting bracket. Use the mounting clamp and capscrew.

Torque Value: 24 n•m [18 ft-lb]

Install the fuel cooler sea water hoses. Tighten the hose clamps.

Torque Value: 3 n•m [30 in-lb]



Finishing Steps

⚠ WARNING ⚠

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⚠ CAUTION ⚠

Failure to properly prime the fuel system on marine engines can result in damage to the fuel cooler. Be sure the drain valve(s) are open. Be sure a good flow of fuel is present through the fuel cooler by cycling the keyswitch at least five times before starting the engine. Refer to Procedure 005-045 in Section 5. (</qs3/pubsys2/xml/en/procedures/40/40-005-045-tr.html>)

⚠ CAUTION ⚠

Use caution when disconnecting or removing fuel lines, replacing filters, and priming the fuel system that fuel is not spilled or drained into the bilge area. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be discarded in accordance with local environmental regulations.

⚠ CAUTION ⚠

Zinc plugs expand and can break off during removal. Inspect the zinc plug to make sure it is in one piece. If not, it must be replaced with a new zinc plug, and the broken pieces must be retrieved from the aftercooler to prevent damage to components downstream in the sea water system.

- Connect the fuel drain line from the fuel cooler to the fuel tanks. See equipment manufacturer service information.

- Connect the fuel drain line from the combined fuel manifold to the fuel cooler at the fuel cooler connection.
- Open the supply and return valves to the engine. See equipment manufacturer service information.
- Install the zinc plug into the aftercooler. Refer to Procedure 008-059 in Section 8 (</qs3/pubsys2/xml/en/procedures/40/40-008-059.html>).
- Open the sea water supply line(s). See equipment manufacturer service information.
- Prime the fuel system. Refer to Procedure 005-045 in Section 5. (</qs3/pubsys2/xml/en/procedures/40/40-005-045-tr.html>)
- Operate the engine. Check for leaks.

Last Modified: 09-Mar-2017
